

LAST NAME: _____

Practice Quiz 03C — MATH 2055 — Fall 2026

Practice for Quiz 03

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Consider the statement

$$\forall x \in \mathbf{R} \exists y \in \mathbf{R} (x < y \wedge y \leq x + 2).$$

- (a) Write its negation using no $\neg$ symbols, and name the logical laws you use.

- (b) Determine whether your negation is true or false, and justify your answer by choosing a suitable y for an arbitrary real x .